

SECDV-FINAL: Security Case Study Report

Your Name
Your Student ID

November 22, 2025

Contents

1	Introduction	2
1.1	Project Overview	2
1.2	Objectives	2
2	System Architecture	2
2.1	Technology Stack	2
2.2	Key Components	2
3	Security Analysis	2
3.1	Authentication & Authorization	2
3.2	Input Validation	2
3.3	Security Vulnerabilities	2
4	Implementation Details	3
5	Testing	3
6	Conclusion	3
7	References	3

1 Introduction

This document presents a security analysis of the SECDV-FINAL project, an event management system built with Node.js and MongoDB.

1.1 Project Overview

Describe your project here - what it does, its main features, architecture, etc.

1.2 Objectives

List the objectives of your case study or project.

2 System Architecture

Describe the system architecture, technology stack, and design decisions.

2.1 Technology Stack

- Backend: Node.js with Express
- Database: MongoDB
- Frontend: Handlebars (HBS), Bootstrap
- Authentication: Session-based

2.2 Key Components

Describe controllers, models, middleware, etc.

3 Security Analysis

Analyze the security features and vulnerabilities of your system.

3.1 Authentication & Authorization

Discuss how authentication and authorization are implemented.

3.2 Input Validation

Discuss input validation mechanisms.

3.3 Security Vulnerabilities

Identify and discuss potential security issues.

4 Implementation Details

Here you can include small, relevant code snippets inline:

```
1 const authenticate = (req, res, next) => {  
2   if (req.session && req.session.userId) {  
3     return next();  
4   }  
5   res.redirect('/login');  
6 };
```

Listing 1: Example Authentication Middleware

5 Testing

Describe your testing approach and results.

6 Conclusion

Summarize your findings and recommendations.

7 References

- OWASP Top 10
- Node.js Security Best Practices
- MongoDB Security Documentation